

Making the Case for Biomarkers in Cancer Treatment



T.J.

Melanoma Survivor, Florida

In August 2012, just weeks after the birth of his second child, 37-year-old T.J. walked into his local emergency room with a spiking fever and did not leave for more than 2 weeks. He was blindsided by the diagnosis he received: metastatic melanoma. Twelve years earlier, a stage IB melanoma had been removed from his chest. T.J. had been careful about his sun exposure ever since. But now, melanoma tumors riddled his lungs, liver, spleen, and small intestine.

T.J.'s first oncologist recommended chemotherapy and expressed doubt that he would be alive in 2 years. T.J. and his wife, Jen, deciding that was not acceptable, sought additional medical opinions and, ultimately settled on a

clinical trial that was testing a combination of immunotherapies that had not previously been given to patients. T.J. did not benefit from this trial. “I was the first to try that combination, and I was the first to fail it,” he said. The regimen was arduous, and he experienced many complications.

Despite this setback, he enrolled in another trial. “I knew that clinical research was far from a guarantee but held the promise of hope,” he said. “My children, Josie and Tommy, were growing up right in front of me, and I was willing to endure anything.” Within a matter of weeks of receiving his first dose of the drug being tested in the new trial—a PD-1 checkpoint inhibitor—his tumors shrank by half. Five years later, T.J. remains cancer free.

Based on the success of this and many other clinical trials, the Food and Drug Administration (FDA) has approved five PD-1/PD-L1 checkpoint inhibitors to treat melanoma and other cancers since 2014.

NCI supported the initial work necessary to unleash the immune system against cancer with studies of ipilimumab (Yervoy®), the first immune checkpoint inhibitor to be approved by FDA. Ipilimumab blocks an immune checkpoint protein called CTLA-4. NCI-funded research had shown that blocking CTLA-4 improved the ability of the immune system to attack cancer in animal models. This pioneering work formed the basis of ipilimumab’s approval for patients with metastatic melanoma in 2011 and paved the way for future immunotherapies like the one that T.J. received.

Long-term follow-up of melanoma patients treated with ipilimumab in clinical trials have demonstrated that approximately 20% of patients were still alive 10 years later. Follow-up data from trials of approved PD-1 inhibitors show that approximately one-third of patients were alive at least 5 years after treatment with nivolumab (Opdivo®), and 40% of patients were alive 3 years after treatment with pembrolizumab (Keytruda®).

For a disease as deadly as metastatic melanoma, this is great progress. Yet, this also means that most patients do not experience a long-lasting benefit from the drugs. “Without further research, we will never know why I am living and so many others have not benefited,” said T.J., reflecting on the need for biomarkers to help doctors identify the best treatment for each cancer patient. Several factors influence the effectiveness of immunotherapy in patients. NCI is supporting ongoing work to identify biomarkers of immunotherapy response, resistance, and toxicity.

It has been 5 years since T.J.’s diagnosis with metastatic melanoma. His aspirations for the next 5 years include watching his son turn 10. “He was just weeks old when my melanoma came back. Every birthday for him is a small milestone for me, too.” Through this journey, T.J. became a blogger and advocate aiming to make a difference in the lives of other patients. “If, for some reason, I am not around in 5 years, I also hope those years are filled with impact so that other patients and families have the most valuable medicine of all—hope.”

This story was originally published in the [NCI Fiscal Year 2019 Annual Plan & Budget Proposal](#). Read the most recent Annual Plan & Budget Proposal at plan.cancer.gov.